

Cluster Analysis on Locally Asymptotically Self-similar Processes

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Plan and Content

- 1 Literature Review: Clustering Stochastic Processes
- 2 Result 1: Dissimilarity Measures
 - Wide-sense Stationary Ergodic Processes
 - Locally Asymptotically Self-similar Processes
- 3 Result 2: Asymptotically Consistent Algorithms
- 4 Result 3: Efficiency Improvement – $\log^*$ -transform
- 5 Result 4: Simulation Study
 - Datasets Preparation and Misclassification Rate
 - Example: Multifractal Brownian motions
- 6 Result 5: Real World – Clustering Global Equity Markets

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Cluster Analysis – K-means method

- **Step 1:** Suggest a fitting dissimilarity measure (e.g. distance).
- **Step 2:** Design a clustering function.

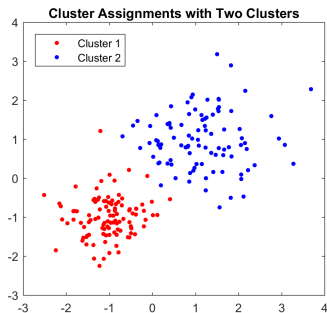
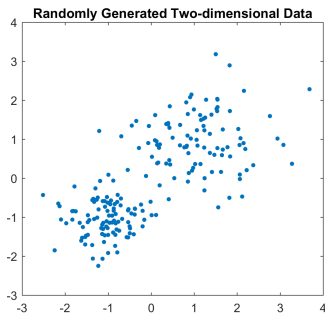


Figure: Euclidean distance.

K -means clustering method ($K = 2$).

Clustering Vectors vs Clustering Stochastic Processes

When using vectors clustering approaches clustering stochastic processes:

- Computationally forbidding $\rightsquigarrow$ huge length.
- Overfitting issue $\rightsquigarrow$ no consideration of path features.

Clustering processes needs novel dissimilarity measures and novel clustering functions.

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Wide-sense Stationary Ergodic Processes (Rao et al. 2019)

We provide *consistent clustering algorithms*:

More General Objects: Wide-sense stationary ergodic processes.

$$\text{Cov}(X_{t_1+h}, \dots, X_{t_n+h}) = \text{Cov}(X_{t_1}, \dots, X_{t_n});$$

$$\nu^*(X_{l\dots n}, m) := \frac{\sum_{i=l}^{n-m+1} X_{i\dots i+m-1}^T X_{i\dots i+m-1}}{n-m-l+2} \xrightarrow[n \rightarrow \infty]{\mathbb{P}} \text{Cov}(X_{l\dots l+m-1}).$$

Covariance-based Dissimilarity: ρ^* is the Frobenius norm.

$$d^*(X^{(1)}, X^{(2)}) := \sum_{m,l=1}^{\infty} w_m w_l \rho^* \left(\text{Cov}(X_{l\dots l+m-1}^{(1)}), \text{Cov}(X_{l\dots l+m-1}^{(2)}) \right)$$

estimated by

$$\widehat{d^*}(x_1, x_2) = \sum_{m=1}^{m_n} \sum_{l=1}^{n-m+1} w_m w_l \rho^* \left(\nu^*(X_{l\dots n}^{(1)}, m), \nu^*(X_{l\dots n}^{(2)}, m) \right).$$

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Locally Asymptotically Self-similar Processes

A continuous-time stochastic process $\{Z_t^{(H(t))}\}_{t \geq 0}$ with its index $H(\bullet)$ being a continuous function valued in $(0, 1)$, is called locally asymptotically self-similar, if for each $t \geq 0$, there exists a non-degenerate self-similar process $\{Y_u^{(H(t))}\}_{u \geq 0}$ with self-similarity index $H(t)$, such that

$$\left\{ \frac{Z_{t+\tau u}^{(H(t+\tau u))} - Z_t^{(H(t))}}{\tau^{H(t)}} \right\}_{u \geq 0} \xrightarrow[\tau \rightarrow 0^+]{f.d.d.} \{Y_u^{(H(t))}\}_{u \geq 0}, \quad (1)$$

where the convergence $\xrightarrow{f.d.d.}$ is in the sense of all the finite dimensional distributions. In (1), $\{Y_u^{(H(t))}\}_u$ is called the *tangent process* of $\{Z_t^{(H(t))}\}_t$ at t .

Locally Asymptotically Self-similar Processes

Assumption ($\mathcal{A}$): The LASSPs' tangent processes' increments are *autocovariance ergodic*.

Proposition: Let $\{Z_t^{(H(t))}\}_{t \geq 0}$ be a locally asymptotically self-similar process satisfying Assumption ($\mathcal{A}$). For each $h > 0$,

$$\left\{ \frac{Z_{t+\tau(u+h)}^{(H(t+\tau(u+h)))} - Z_{t+\tau u}^{(H(t+\tau u))}}{\tau^{H(t)}} \right\}_{u \geq 0} \xrightarrow[\tau \rightarrow 0^+]{\text{f.d.d.}} \{X_{u,h}^{(H(t))}\}_{u \geq 0}, \quad (2)$$

where $\{X_{u,h}^{(H(t))}\}_{u \geq 0} := \{Y_{u+h}^{(H(t))} - Y_u^{(H(t))}\}_{u \geq 0}$ is an autocovariance ergodic process.

Proof

Proof.

Let's fix $h > 0$ and pick any finite discrete time indexes set $T \subset [0, +\infty)$. Under Assumption (A), the f.d.d. convergence (1) holds. It then implies

$$\left(\frac{Z_{t+\tau(u+h)}^{(H(t+\tau(u+h)))} - Z_t^{(H(t))}}{\tau^{H(t)}}, \frac{Z_{t+\tau u}^{(H(t+\tau u))} - Z_t^{(H(t))}}{\tau^{H(t)}} \right)_{u \in T} \xrightarrow[\tau \rightarrow 0^+]{\text{law}} (Y_{u+h}^{(H(t))}, Y_u^{(H(t))})_{u \in T}, \quad (3)$$

It follows from (3) and the continuous mapping theorem that

$$\left(\frac{Z_{t+\tau(u+h)}^{(H(t+\tau(u+h)))} - Z_t^{(H(t))}}{\tau^{H(t)}} - \frac{Z_{t+\tau u}^{(H(t+\tau u))} - Z_t^{(H(t))}}{\tau^{H(t)}} \right)_{u \in T} \xrightarrow[\tau \rightarrow 0^+]{\text{law}} (Y_{u+h}^{(H(t))} - Y_u^{(H(t))})_{u \in T}. \quad (4)$$



Locally Asymptotically Self-similar Processes

Equation(2) can be interpreted as: when τ is sufficiently small,

$$\left\{ Z_{t+\tau(u+h)}^{(H(t+\tau(u+h)))} - Z_{t+\tau u}^{(H(t+\tau u))} \right\}_{u \in [0, Kh]} \stackrel{\text{f.d.d.}}{\approx} \left\{ \tau^{H(t)} X_{u,h}^{(H(t))} \right\}_{u \in [0, Kh]}, \quad (5)$$

where K is an arbitrary positive integer. Smaller is τ , closer are the two hand sides of (5).

Covariance-based Dissimilarity Measure

Given a path $Z_{t_1}^{(H(t_1))}, \dots, Z_{t_n}^{(H(t_n))}$ sampled from $\{Z_t^{(H(t))}\}$, for $i \in \{1, \dots, n - K - 1\}$, the path

$$z^{(i)} := \left(Z_{t_{i+1}}^{(H(t_{i+1}))} - Z_{t_i}^{(H(t_i))}, \dots, Z_{t_{i+1+K}}^{(H(t_{i+1+K}))} - Z_{t_{i+K}}^{(H(t_{i+K}))} \right), \quad (6)$$

is approximately distributed as a wide-sense stationary ergodic process, with self-similar index $H(i)$. Therefore we define the empirical covariance-based dissimilarity measure between 2 paths of locally asymptotically self-similar processes z_1 and z_2 as

$$\widehat{d^*}(z_1, z_2) := \frac{1}{n - K - 1} \sum_{i=1}^{n-K-1} \widehat{d^*}(z_1^{(i)}, z_2^{(i)}). \quad (7)$$

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Ground Truth of Covariance Structure

Definition (Ground-truth G of covariance structures)

Let

$$G = \{G_1, \dots, G_\kappa\}$$

be a partitioning of $\mathbb{N}$ into κ disjoint sets G_k , $k = 1, \dots, \kappa$, such that the means and covariance structures of x_i , $i \in \mathbb{N}$ are identical, if and only if $i \in G_k$ for some $k = 1, \dots, \kappa$. Such G is called ground-truth. We also denote by $G|_N$ the restriction of G to the first N sequences:

$$G|_N = \{G_k \cap \{1, \dots, N\} : k = 1, \dots, \kappa\}.$$

Information Collection: Offline Setting

The observations are assumed to be a fixed finite number N of paths:

$$\mathbf{x}_1 = (X_1^{(1)}, \dots, X_{n_1}^{(1)}), \dots, \mathbf{x}_N = (X_1^{(N)}, \dots, X_{n_N}^{(N)}).$$

Definition (Consistency: offline setting)

A clustering function f is consistent for a set of sequences S if $f(S, \kappa) = G$. Denoting $n = \min\{n_1, \dots, n_N\}$, f is called strongly asymptotically consistent in the offline sense if:

$$\mathbb{P} \left(\lim_{n \rightarrow \infty} f(S, \kappa) = G \right) = 1.$$

It is weakly asymptotically consistent if $\lim_{n \rightarrow \infty} \mathbb{P}(f(S, \kappa) = G) = 1$.

Information Collection: Online Setting

The observations, having growing length and number of scenarios with respect to time t , are denoted by

$$\mathbf{x}_1 = \left(X_1^{(1)}, \dots, X_{n_1}^{(1)} \right), \dots, \mathbf{x}_{N(t)} = \left(X_1^{(N(t))}, \dots, X_{n_{N(t)}}^{(N(t))} \right).$$

Definition (Consistency: online setting)

A clustering function is strongly (RESP. weakly) asymptotically consistent in the online sense, if for every $N \in \mathbb{N}$ the clustering $f(S(t), \kappa)|_N$ is strongly (RESP. weakly) asymptotically consistent in the offline sense, where $f(S(t), \kappa)|_N$ is the clustering $f(S(t), \kappa)$ restricted to the first N sequences:

$$f(S(t), \kappa)|_N = \{f(S(t), \kappa) \cap \{1, \dots, N\} : k = 1, \dots, \kappa\}.$$

Offline Algorithm

Algorithm 1: Offline clustering, with known κ

Input: *sample sequences* $S = \{x_1, \dots, x_N\}$; *number* κ *of clusters.*

1 $(c_1, c_2) \leftarrow \underset{(i,j) \in \{1, \dots, N\}^2, i < j}{\operatorname{argmax}} \widehat{d}^*(x_i, x_j);$

2 $C_1 \leftarrow \{c_1\}; C_2 \leftarrow \{c_2\};$

3 **for** $k = 3, \dots, \kappa$ **do**

4 $c_k \leftarrow \underset{i=1, \dots, N}{\operatorname{argmax}} \min_{j=1, \dots, k-1} \widehat{d}^*(x_i, x_{c_j})$

5 **end**

6 *Assign the remaining points to closest centers:* **for** $i = 1, \dots, N$ **do**

7 $k \leftarrow \underset{k \in \{1, \dots, \kappa\}}{\operatorname{argmin}} \left\{ \widehat{d}^*(x_i, x_j) : j \in C_k \right\};$

8 $C_k \leftarrow C_k \cup \{i\}$

9 **end**

Output: The κ clusters $\{C_1, C_2, \dots, C_\kappa\}$.

Online Algorithm

Algorithm 2: Online clustering, with known κ

Input: number κ of target clusters.

```

1  for  $t = 1, \dots, \infty$  do
2      Obtain new sequences:  $S(t) \leftarrow \{x_1^t, \dots, x_{N(t)}^t\}$ ;
3      Initialize the normalization factor:  $\eta \leftarrow 0$ ;
4      Initialize the final clusters:  $C_k(t) \leftarrow \emptyset, k = 1, \dots, \kappa$ ;
5      Generate  $N(t) - \kappa + 1$  candidate cluster centers:
6      for  $j = \kappa, \dots, N(t)$  do
7           $\{C_1^j, \dots, C_\kappa^j\} \leftarrow \text{Alg1}(\{x_1^t, \dots, x_j^t\}, \kappa)$ ;
8           $(c_1^j, \dots, c_\kappa^j) \leftarrow \text{sort}(\min \{i \in C_k^j\}, k = 1, \dots, \kappa)$ ;
9           $\gamma_j \leftarrow \min_{k, k' \in \{1, \dots, \kappa\}, k \neq k'} \widehat{d}^* \left( x_{c_k^j}^t, x_{c_{k'}^j}^t \right)$ ;
10          $\omega_j \leftarrow 1/j(j+1)$ ;
11          $\eta \leftarrow \eta + \omega_j \gamma_j$ 
12     end
13     Assign each point to a cluster:
14     for  $i = 1, \dots, N(t)$  do
15          $k \leftarrow \operatorname{argmin}_{k' \in \{1, \dots, \kappa\}} \frac{1}{\eta} \sum_{j=\kappa}^{N(t)} w_j \gamma_j \widehat{d}^* (x_i^t, x_{c_{k'}^j}^t)$ ;
16          $C_k(t) \leftarrow C_k(t) \cup \{i\}$ 
17     end
18 end

```

Output: The κ clusters $\{C_1(t), \dots, C_\kappa(t)\}, t = 1, 2, \dots, \infty$.

Approximate Asymptotic Consistency

Fix K . Let $Z^{(1)}, Z^{(2)}$ be 2 LASSPs with functional indexes H_1, H_2 . Let $(z_1^{(1)}, \dots, z_1^{(n-K-1)}), (z_2^{(1)}, \dots, z_2^{(n-K-1)})$ be respectively their sample paths z_1, z_2 ' increments. For $j = 1, 2$, define the normalized increments by:

$$\mathcal{H}(z_j^{(i)}) := \frac{z_j^{(i)}}{\Delta t^{H_j(t_i)}}, \text{ for } i = 1, \dots, n - K - 1. \quad (8)$$

Fix $L \geq 1$. For each $\widehat{d}^*(z_1, z_2)$, define

$$\widetilde{d}^*(z_1, z_2) := \frac{1}{L} \sum_{i=1}^L \widehat{d} \left(\mathcal{H}(z_1^{(i)}), \mathcal{H}(z_2^{(i)}) \right). \quad (9)$$

Proposition: For any N independent sample paths $z_1, z_2, \dots, z_N$,

$$\left(\widetilde{d}^*(z_i, z_j) \right)_{i,j \in \{1, \dots, N\}, i \neq j} \xrightarrow[\Delta t \rightarrow 0]{\text{law}} \left(\widehat{d}^*(x_i, x_j) \right)_{i,j \in \{1, \dots, N\}, i \neq j}, \quad (10)$$

where x_1, x_2 are the increments of the tangent processes corresponding to z_1, z_2 respectively.

Theorem of Approximate Asymptotic Consistency

Theorem

Under Assumption (A), Algorithm 1 is approximately asymptotically consistent for clustering the paths $S = \{z_1, \dots, z_N\}$. This means: if $\widehat{d}^$ is replaced with $\widetilde{d}^*$ in Algorithm 1, the output clusters converge to the ground truths of the increments of the corresponding tangent processes $S' = \{x_1, \dots, x_N\}$, as $\Delta t \rightarrow 0$ and $n_{\min} := \min\{n_1, \dots, n_N\} \rightarrow \infty$. More formally,*

$$\lim_{n_{\min} \rightarrow +\infty} \lim_{\Delta t \rightarrow 0} \mathbb{P} \left(f(S, \kappa, \widetilde{d}^*) = G_{S'} \right) = 1, \quad (11)$$

where $G_{S'}$ denotes the ground truths of the covariance structures that generate the set of paths S' .

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Efficient Dissimilarity Measure

Definition (Efficient dissimilarity measure)

Assume that the samples $S = \{x(\xi) : \xi \in \mathcal{H}\}$ ($\mathcal{H} \subset \mathbb{R}^q$ for some $q \in \mathbb{N}$), meaning that all the paths $x(\xi)$ are indexed by a set of real-valued parameters ξ . Then a clustering function is called efficient if its dissimilarity measure d satisfies that, there exists $c > 0$ so that for any $x(\xi_1), x(\xi_2) \in S$,

$$d(x(\xi_1), x(\xi_2)) = c \|\xi_1 - \xi_2\|,$$

where $\|\cdot\|$ denotes some norm defined over $\mathbb{R}^q$.

Counter-example

Since the increment process $\{Z_h^{(H)}(t)\}$ of fractional Brownian motion satisfies $\text{Cov}(Z_h^{(H)}(s), Z_h^{(H)}(t)) \sim |s - t|^{2H}$, h small, then for $\{Z_h^{(H_1)}(t)\}$ and $\{Z_h^{(H_2)}(t)\}$, its dissimilarity measure $d^*(Z_h^{(H_1)}, Z_h^{(H_2)})$ is not efficient.

Motivations for $\log^*$ -transformation

Definition ($\log^*$ -transformation)

Define the $\log^*$ -transformation to be: for $x \in \mathbb{R}$,

$$\log^*(x) := \text{sgn}(x) \log |x| = \begin{cases} \log(x) & \text{if } x > 0; \\ -\log(-x) & \text{if } x < 0; \\ 0 & \text{if } x = 0. \end{cases}$$

Motivation 1:

The $\log^*$ function transforms the current dissimilarity measure to the one which “linearly” depends on its variable H .

Motivation 2:

The value $\log^*(x)$ preserves the sign of x , which leads to the consequence that large distance between x, y implies large distance between $\log^*(x)$ and $\log^*(y)$.

An Empirical Covariance-based Dissimilarity Measure

A new empirical covariance-based dissimilarity measure with $\log^*$ -transformation:

$$\widehat{d}^{**}(z_1, z_2) := \sum_{m=1}^{m_n} \sum_{l=1}^{n-m+1} w_m w_l \rho^* \left(\nu^{**}(Z_{l\dots n}^{(H_1)}, m), \nu^{**}(Z_{l\dots n}^{(H_2)}, m) \right),$$

where $\nu^{**}(Z_{l\dots n}^{(H_1)}, m)$ is the empirical covariance matrix of $Z_h^{(H_1)}$, $\nu^*(Z_{l\dots n}^{(H_1)}, m)$, with each of its coefficients transformed by $\log^*$: let $M = \{M_{ij}\}_{i=1,\dots,m; j=1,\dots,n}$ be an arbitrary real-valued matrix, define

$$\log^* M := \{\log^* M_{ij}\}_{i=1,\dots,m; j=1,\dots,n}.$$

Then we have

$$\nu^{**}(Z_{l\dots n}^{(H_1)}, m) := \log^* \left(\nu^*(Z_{l\dots n}^{(H_1)}, m) \right).$$

Application: Self-similar Processes

Definition (Self-similar process)

A process $X^{(H)} = \{X_t^{(H)}\}_{t \geq 0}$ is self-similar with index $H > 0$ if, for all $n \in \mathbb{N}$, all $t_1, \dots, t_n \geq 0$, and all $c > 0$,

$$\left(X_{ct_1}^{(H)}, \dots, X_{ct_n}^{(H)}\right) \stackrel{\text{law}}{=} \left(c^H X_{t_1}^{(H)}, \dots, c^H X_{t_n}^{(H)}\right).$$

Theorem (Relationship to wide-sense stationary ergodic processes)

Let $\{X_t^{(H)}\}_{t \geq 0}$ be a zero-mean self-similar process with index H and $\mathbb{E}|X_1^{(H)}|^2 < +\infty$. Then it has wide-sense stationary ergodic increments *if and only if* for any $s, t \geq 0$,

$$\text{Cov}\left(X_s^{(H)}, X_t^{(H)}\right) = \frac{\mathbb{E}|X_1^{(H)}|^2}{2} \left(|s|^{2H} + |t|^{2H} - |s - t|^{2H}\right).$$

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- Offline data preparation.

$$\implies \{X^{\alpha_i}(t)\}, i = 1, \dots, 5 \text{ for distinct } \alpha_1, \dots, \alpha_5.$$

$\Rightarrow$ Total number of paths $N = 50$.

$$\Rightarrow t = \frac{3}{n}, \frac{6}{n}, \dots, \frac{n-3}{n}, 1, \text{ with } n = 150.$$


Online Data Preparation

- Online data preparation.

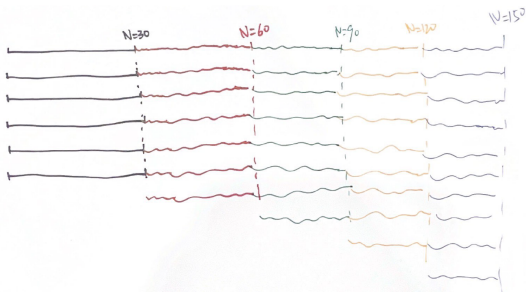
(i) Number of clusters $\kappa = 5$:

$\Rightarrow \{X^{\alpha_i}(t)\}$, $i = 1, \dots, 5$ for distinct $\alpha_1, \dots, \alpha_5$.

(ii) Each cluster contains $R(t)$ paths at time t .

$\Rightarrow R(3/n) = 6, R(33/n) = 7, R(63/n) = 8, R(93/n) = 9, R(123) = 10$, with $n = 150$.

(iii) Each newly added path starts with length 3, after each $3/n$ ($n = 150$) time period its length adds 3.



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Multifractional Brownian Motions $\{W^{(H(t))}(t)\}$

The multifractional Brownian motion (mBm) $\{W^{H(t)}(t)\}_{t \geq 0}$ is a zero-mean continuous-time Gaussian process, which can be defined via its covariance function: for $s, t \geq 0$,

$$\begin{aligned} \text{Cov}(W^{H(t)}(t), W^{H(s)}(s)) &= D(H(t), H(s)) \\ &\times (t^{H(t)+H(s)} + s^{H(t)+H(s)} - |t-s|^{H(t)+H(s)}), \end{aligned} \quad (12)$$

where

$$D(t, s) := \frac{\sqrt{\Gamma(2t+1)\Gamma(2s+1)\sin(\pi t)\sin(\pi s)}}{2\Gamma(t+s+1)\sin(\pi(t+s)/2)}.$$

Its tangent process at t is an fBm $\{B^{(H(t))}(u)\}_u$:

$$\left\{ \frac{W^{H(t+\tau u)}(t+\tau u) - W^{H(t)}(t)}{\tau^{H(t)}} \right\}_u \xrightarrow[\tau \rightarrow 0^+]{\text{f.d.d.}} C_{H(t)} \left\{ B^{(H(t))}(u) \right\}_u, \quad (13)$$

where $C_{H(t)}$ is a deterministic function only depending on $H(t)$.

Result: Multifractional Brownian Motions

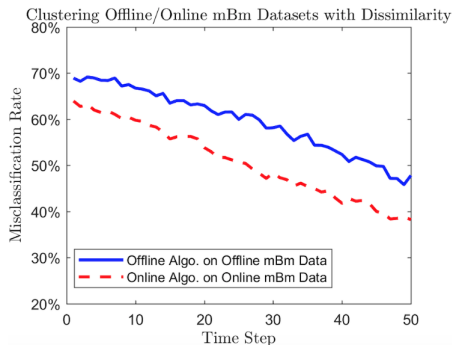
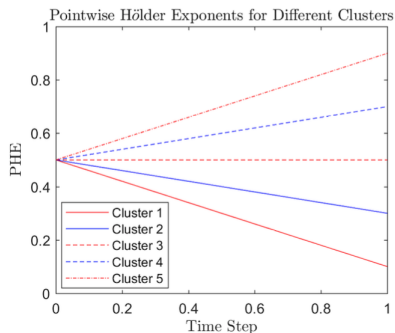


Figure: The graphs compare the misclassification rates of offline (online) Algorithm applied to offline (online) dataset (with $\log^*$ transformation) of mBm. 100 runs are performed at each time step t to compute the misclassification rate.

Result: Multifractional Brownian Motions

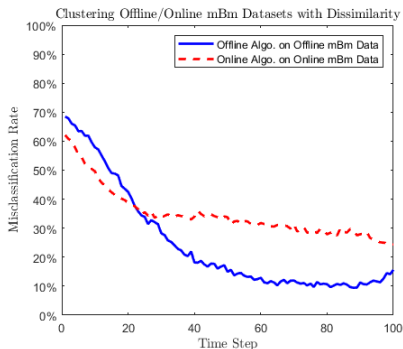
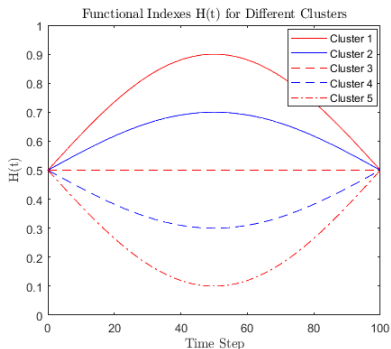


Figure: The graphs compare the misclassification rates of offline (online) Algorithm applied to offline (online) dataset (with $\log^*$ transformation) of mBm. 100 runs are performed at each time step t to compute the misclassification rate.

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Real World: Clustering Global Equity Markets

- Step 1:** Collect data: we use two asset classes from global financial markets: the country-level stock index(MSCI ACWI (All Country World Index)) and the sovereign credit default swap spread.
- Step 2:** Assume it is some multifractional Brownian motion.
- Step 3:** We compare the clustering results using offline and online datasets with the number of clusters based on
- (i) geographical regions (4 groups: Americas, Europe & Middle East, Pacific and Asia)
 - (ii) development level (2 groups: emerging markets and developed markets).

Real World: Clustering Results

Table: The misclassification rates of clustering algorithms on different datasets. The clustering facts are geographical region and development level, respectively. Panel A presents the results from clustering global stock indexes, and Panel B presents the results from clustering sovereign CDS spreads.

Panel A	Offline Algorithm		Online Algorithm	
	Regions	Emerging/Developed	Regions	Emerging/Developed
Stock Returns				
offline dataset	61.70%	29.79%	55.32%	36.17%
online dataset	53.19%	44.68%	51.06%	14.89%

Panel B	Offline Algorithm		Online Algorithm	
	Regions	Emerging/Developed	Regions	Emerging/Developed
CDS Spreads				
offline dataset	64.29%	28.57%	71.43%	26.19%
online dataset	54.76%	47.62%	59.52%	26.19%

Real World: Clustering Results

Pabel A: Equity Indexes Returns

Group 1 (Emerging Markets)		Group 2 (Developed Markets)	
Incorrect - Offline	Incorrect - Online	Incorrect - Offline	Incorrect - Online
Austria	Austria	Korea	Czech Republic
Finland	Finland	Chile	Qatar
Germany	Portugal	Philippines	Peru
Ireland		Malaysia	South Africa
Italy		Mexico	
Norway			
Portugal			
Spain			
New Zealand			

Pabel B: Sovereign CDS Spreads

Group 1 (Emerging Markets)		Group 2 (Developed Markets)	
Incorrect - Offline	Incorrect - Online	Incorrect - Offline	Incorrect - Online
Ireland	Ireland	Chile	Chile
Italy	Portugal	China (Mainland)	China (Mainland)
Portugal		Czech Republic	Czech Republic
Spain		Korea	Hungary
		Malaysia	Korea
		Mexico	Malaysia
		Poland	Mexico
		Thailand	Poland
			Thailand

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